

torch.atan2#

<https://pytorch.org/docs/stable/generated/torch.atan2.html#torch.atan2>

Description:

Element-wise arctangent of $\frac{\text{input}_i}{\text{other}_i}$ / $\frac{\text{other}_i}{\text{input}_i}$ with consideration of the quadrant. Returns a new tensor with the signed angles in radians between vector $(\text{other}_i, \text{input}_i)$ and vector $(1, 0)$. (Note that other_i , the second parameter, is the x-coordinate, while input_i , the first parameter, is the y-coordinate.) The shapes of input and other must be broadcastable.

input (Tensor) – the first input tensor
other (Tensor) – the second input tensor
out (Tensor, optional) – the output tensor.

Function Signature

```
torch.atan2(input: Tensor, other: Tensor, *, out: Optional[Tensor]) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(4)
>>> a
tensor([ 0.9041,  0.0196, -0.3108, -2.4423])
>>> torch.atan2(a, torch.randn(4))
tensor([ 0.9833,  0.0811, -1.9743, -1.4151])
```

Detailed Documentation

Documentation

Element-wise arctangent of $\text{input}_i / \text{other}_i$ with consideration of the quadrant. Returns a new tensor with the signed angles in radians between vector $(\text{other}_i, \text{input}_i)$ and vector $(1, 0)$. (Note that other_i , the second parameter, is the x-coordinate, while input_i , the first parameter, is the y-coordinate.)

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